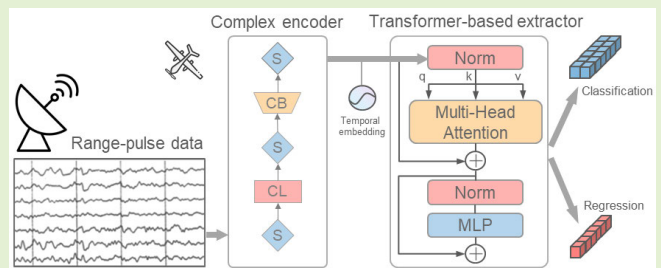


Echoformer: Transformer Architecture Based on Radar Echo Characteristics for UAV Detection

Yipu Yang, Fan Yang[✉], Liguo Sun, Ti Xiang, and Pin Lv

Abstract—While recent years have witnessed an increasing number of commercial applications of unmanned aerial vehicles (UAVs), an imperative problem people have to face is the rapid growth of malicious use. So, it is imperative for security agencies to develop anti-UAV technology. The introduction of deep learning (DL) has a positive influence on radar signal processing, but DL-based methodologies have yet to be widespread in radar target detection because of the lack of unique architecture based on radar echo characteristics and the annotation method of radar data. In this article, a novel Transformer-based architecture is proposed, which transforms the problem of UAV detection into a binary classification task in each range cell. The complex encoder architecture and the Transformer-based extractor are designed to extract the Doppler frequency shift feature and the micro-Doppler signature (mDS) of a UAV simultaneously. The well-designed architecture based on radar echo characteristics can achieve a combination training of echoes with different coherent processing intervals (CPIs). In addition, we provide an annotation method and a data augmentation skill for our real measured dataset. The results of the experiment demonstrate that the proposed method has better detection performance and measuring accuracy under different SNRs in comparison with traditional radar target detection and other DL-based methods.

Index Terms—Doppler frequency shift, micro-Doppler signature (mDS), radar echo processing, Transformer, unmanned aerial vehicle (UAV) detection.



I. INTRODUCTION

UNMANNED aerial vehicles, commonly known as drones, have been widely used in the field of the military over the past two decades [1]. With the increase in the number of manufacturers and the decline of production costs, recent years have witnessed the soaring civil application of unmanned aerial vehicles (UAVs) [2], which has a tremendous achievement in commercial applications, agriculture, and transportation. It also provides vast support for such tasks as emergency item deliveries, emergency rescue in the case of disaster, and visual

surveillance for crowd safety [3]. However, this fact also gives rise to malicious use, such as UAV intrusion in the airport, privacy violations, and even terrorist attacks [4]. Therefore, equipping necessary sensors that can detect UAVs for security, police, aviation, and other agencies is indispensable for air safety and prohibiting illegal flying [5].

The demands to protect critical infrastructures from the above threats and ensure the legitimate use of UAVs have brought advances in counter-UAV (c-UAV) applications. Nowadays, c-UAV systems comprise a multisensory arsenal often including electro-optical, thermal, acoustic, radio frequency sensors, and radar [6]. Electro-optical sensors, a kind of passive detector, are widely applied in various object detection tasks. The rich textures and structural features of objects are presented in the visual images, which contributes to acquiring an ideal performance of the nearby drones' detection. Unfortunately, detection performance is poor at night and in the fog where the light is too weak to obtain high-quality images [7]. In addition, too small a size for distant targets in the images probably results in the failure of the detection as well [8]. RF sensors are widely applied to various systems owing to their simplicity and low computation [9], yet the problems of undesired target detection reliability, high false alarm probability, and the lack of position information are faced simultaneously [10]. Using these sensors only judges if the targets are present or absent when scanning the suspicious

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regions. It is inaccessible for us to obtain essential state data, like the distance and velocity of the flying UAVs. However, radar, widely in perceptual systems, is a type of active sensor, which is in a position to compensate above drawbacks. Consequently, radar is very competent in long-range detection and able to work reliably all day and in all weather, even in the case of poor weather and light [11], [12].

Typically, the difference in the Doppler frequency shift caused by the velocity of objects can help us distinguish motion targets from ground clutter [9], [13], but the target signal in the echo is too weak to be separated from the clutter sometimes on account of the small radar cross section (RCS) and low flying altitude [14], [15]. In addition, it is difficult for us to detect drones with low flying speeds from the background because of the small Doppler frequency shift. Therefore, adaptive and stable detection of low-observable UAVs by pulse-Doppler radar has been a challenging and hot problem in the detection and measurement of radar signals.

Nowadays, an increasing number of researchers resort to DL-based methods to detect radar targets, in spite of radar target detection being a classic and traditional research area. However, deep learning (DL) has yet to take over for tasks concerning radar data [11]. As far as I know, it is indispensable for the training of the end-to-end DL pipelines to annotate plenty of ground-truth data, but the annotation of radar data can only be performed by experts in the field. Only obtaining raw data from a radar may not be enough because radar raw signal is the rectangular form of complex numbers, so scholars usually modify the data representation for the purpose of sharpening spatiotemporal and intrinsic information. Radar images as the most common representation of radar signals are input into the deep neural networks (DNNs), which is a popular solution at present. The DL-based UAV detection methods essentially are divided into two categories: one is based on object detection in the field of computer vision (CV), and the other comes from the classification tasks: the range-Doppler fragment obtained through a sliding window can be classified into two categories, which determines whether the target is present or not in the fragment. However, the drawbacks caused by the format of the input are obvious: 1) the format of radar image drops out the intrinsic information of signals, such as the phase; 2) in the process of deployment, a fixed-size window sliding over the whole range-Doppler cells is used to search the position of targets whose positioning error is likely to have an adverse influence on the result; and 3) the fixed-size window limits the detection range, which merely is several range cells.

To make up for the deficiencies in previous works, a novel Transformer-based architecture that can simultaneously extract the Doppler frequency shift feature and micro-Doppler signature (mDS) is proposed in this article. Radar raw data after matched filtering (MF) is directly taken as the input, instead of range-Doppler maps after coherent integration. Coherent integration is applied over multiple pulses by applying a fast Fourier transform (FFT) over the whole Doppler cells in each range cell, the process of which is linear computation. The proposed method replaces it with the DNN, since the DNN consists of multiple linear and nonlinear functions, which can make a more comprehensive description and analysis of the

features of radar raw echo. In light of the fact that radar echo is intrinsically temporal data, a novel Transformer-based backbone is adopted to extract the features of radar echo. To realize large-range detection and improve the availability of our algorithm, the problem of UAV detection is transformed into the classification task of the range cells. Not only can the proposed method focus on the Doppler frequency shift of the moving target, but also extract the mDS of the target over the slow-time domain. Concretely, the contributions are fourfold.

- 1) Instead of pulse cancellation, coherent integration, and other clutter suppression processes in the previous work, an end-to-end trainable pipeline that directly takes the raw pulse-range data after MF as the input is designed, as these processes can be seen as the combination of the convolution operation.
- 2) To realize large-range detection and simultaneously extract Doppler frequency shift and mDS, the UAV detection is transformed into a binary classification task to decide whether the target is present in each range cell or not.
- 3) Considering radar echo is complex numbers in both fast- and slow-time domains, complex encoder and Transformer-based feature extractor are presented to achieve combination training of pulse-range maps with different coherent processing interval (CPI).
- 4) The main reasons why DL-based methods have yet been widespread in tasks concerning radar data is that the annotation of radar data demands professional knowledge in the field of radar and there is a lack of data augmentation and other training skills about radar echoes. To settle these problems, sample frequency shift augmentation (SFSA) is proposed to generate target signals, which can improve generalization. And the presented method can convert UAV fight state information (velocity and distance) into trainable labels available for DL-based algorithms. In addition, a real measured dataset collected with a pulse-Doppler radar is built to facilitate the development of DL-based methods for UAV detection.

The rest of the article is organized as follows. Section II gives the relevant literature on UAV detection. Radar parameters analysis and problem formulation are introduced in Section III. In Section IV, this article describes the main UAV detection algorithm of our work. The ways of the annotation of radar data and expanding dataset samples are presented in Section V. The experimental results of the proposed method and performance evaluation are presented in Section VI. Conclusion is drawn in Section VII.

II. RELATED WORK

A. Traditional Methods for UAV Detection

In earlier research, one of the representative traditional methods is essentially based on the constant false alarm rate (CFAR) processor after the clutter cancellation and its improved variants [16]. UAV detection with radar is mainly achieved with the classic radar signal processing for target

detection using Doppler processing and hypothesis testing under the CFAR algorithm as it is described in [17]. The cell averaging CFA (CA-CFAR) detector is the most effective and widespread method, which compares the power of the cell under test (CUT) with the threshold in reference cells to decide whether the target is present or not [18]. However, the threshold sensitivity is an intractable problem, where the small and weak targets are likely to be ignored easily. To solve this question, [19] proposed the adaptive CFAR, which can estimate clutter distribution in a valid way. The case that a reference window is close to clutter edges may contribute to the situations of false alarms or miss alarms, especially causing the target masking in multitarget scenes, though it has a better performance in homogeneous environments. The ordered-statistic CFAR (OS-CFAR) optimizes the result of radar detection based on obtaining the threshold by sorting data samples. So, it has a better performance in clutter edges and multitarget scenes [20]. A generalized likelihood ratio test (GLRT) detector is proposed instructively by Kelly, which promotes the rapid development of radar target detection [21]. With the study of the clutter model, the original Gaussian model is no longer applicable to some nonuniform clutter cases, so an adaptive normalized matched filter (ANMF) is designed, which makes the detector have CFAR properties, to detect moving targets in sea clutter [22].

Another impressive group of UAV detection methods is built on the mDS differences. In the spectrogram, besides the echo caused by the UAV itself motion pose, the micromotions of vibrating (by engines or motors) and rotating structures (by propellers) of UAVs contribute to the unique mDS, which can provide decision features for the detecting UAVs [23], especially in the wideband radar. De Wit et al. [24] first utilized the m-D signature for UAV classification and produced the mDS with spectrogram [short time Fourier transform (STFT)] and with the ceprogram in [25]. They extracted the mDS to generate key characteristics from the radar echo such as rotation rate, blade tip velocity, rotor diameter, and the number of rotors, to classify between different rotary-wing type UAVs. To better model radar data and tackle the curse of dimensionality, [26] proposed a three-step classification framework for UAV detection. The greedy subspace clustering, multi-Gaussian subspace analysis, and a subspace-fusion scheme were designed to tackle the problems discussed above. Some scholars focused their study on automatically classifying UAVs by combining the classifier in machine learning (ML) with mDS extraction. Some mDS-based methods can classify targets in detail, but using radar data fragments obtained through a sliding window limits the detection range and is not suited for multitarget detection [5]. In addition, it is necessary to formulate objects' rotation and motion by mathematic models, which need to be matched with the motion characteristics of the target to improve the signal gain, if we would like to extract the mDS feature by hand.

B. DL-Based Methods for UAV Detection

Recently, DL-based methods have made great achievements in CV, natural language processing (NLP), and some other tasks [16], [27]. Thus, some researchers attempted to detect

and classify the UAVs using DL-based algorithms. Kim's work is the first one that utilizes convolution neural networks (CNNs) to classify UAVs from the mDS spectrograms [28]. The authors adopted GoogleNet and trained it with spectrograms from a frequency-modulated continuous wave (FMCW) radar. Since radar echo is temporal data, Sun et al. [23] presented an mDS-based surveillance method by applying short-term memory neural network (LSTM), which exploited the hidden time relationship from radar data but the extraction of feature used traditional methods, such as principal component analysis (PCA) and STFT. Brooks et al. [29] introduced an expressive simulator for blade-propelled engines such as drones with sufficiently realistic variational characteristics, but it experiments on simulated data. A persistent range-Doppler radar was used for target detection and recognition [4], where a CNN has been implemented to extract high-level interpretations (target type and location), which takes as input arrays of range-Doppler radar data, and predicts their class (car, person, or drone). Although the dataset that they used is publicly available in Kaggle to conduct research, the dataset was constructed by a sliding window on the range-Doppler maps, which cannot perform large-range detection. Wang et al. [30] transformed the target detection problem into a binary classification between the target and clutter classes where the fixed size window slides over the complete range-Doppler map so that all range-Doppler cells are checked. Wang et al. [5] proposed a CNN with two heads: one for the classification of the input range-Doppler map patch into target present or target absent and another for the regression of offset between the target and the patch center. The NMS mechanism based on location voting with triple filtering is developed to effectively reduce false alarms.

In general, according to the different tasks of the model, the DL-based UAV detection methods broadly fit into two categories: one is based on object detection, which is inspired by image recognition in CV. YOLO, SSD, and Faster RCNN are all popular and excellent object recognition algorithms today [31]. Extracting object features by a backbone, these algorithms can recognize what the object is and where it is in an image by taking a 2-D bounding box including category information as ground truth. In the field of UAV detection, these kinds of solutions take rang-Doppler maps as the input of the network [32]. The merits of them are using universal and well-applied pipelines, easy target annotation (bright area on range-Doppler image), and the availability of large-range detection (multiple range cells and Doppler cells). However, the case of heavy clutter background and target close to zero frequency may lead to a large number of false alarms, which has a negative impact on target detection in the process of annotating ground truth. Moreover, we reckon that using a bounding box to annotate the target area with strong energy accumulation merely includes the features of the Doppler frequency of targets. The mDS in the whole Doppler domain is dropped out, which is the main reason for the above shortcomings.

The other methods are based on the classification: the spectrogram over time or range-Doppler fragment obtained through a sliding window will be classified into two categories, which

TABLE I

RADAR PARAMETERS. THE ANALYSIS OF PERFORMANCE PARAMETERS IS INDISPENSABLE TO TARGET ANNOTATION IN (20) AND (21)

Items	Performance Parameters
carrier frequency (f_c)	35 GHz
PRF (f_r)	32 KHz
bandwidth (B)	80 MHz
PRT (T_r)	31.25 μ s
wavelength (λ)	0.0086 m
range resolution (Δr)	1.875 m
$T_{cpi}=10$ ms	0.429 m/s
velocity resolution (Δv)	0.286 m/s
$T_{cpi}=15$ ms	0.214 m/s
$T_{cpi}=20$ ms	0.214 m/s
minimum radar range (r_{min})	1.875 m
maximum unambiguous range (r_{max})	4688 m
maximum unambiguous velocity (v_{max})	137.14 m/s

determines whether the target is present or not in the input data. The fixed-size data fragment is well suited to extract features with CNNs and other neural networks where the difference of features between various things and clutter background can be distinguished in detail [11], [30]. In contrast to approaches based on object detection, it is good at extracting the hidden mDS generated by the target itself pose changes and micromotions, so multiclass classification can be finished well [33]. Le et al. [34] presented the inception-residual neural network (IRNN) for target classification based on micro-Doppler radar image data. The well-designed IRNN structure is analyzed by tuning the hyperparameters to seek an equilibrium between computational complexity and accuracy. Zhu et al. [35] focused on the recognition of UAVs under low-SNR scenes. A deep convolution denoising encoder is designed to remove the noise without suppressing mDS.

III. RADAR PERFORMANCE ANALYSIS AND PROBLEM FORMULATION

A. Pulse-Doppler Radar Parameters

In this study, radar whose multiple coherent pulses are transmitted works in pulse-Doppler mode and then the received signals are further processed as discrete digital signals after the time sampling. Eventually, the discrete pulse sequence including echoes is obtained after the MF. Radar performance makes a huge difference to the suitability of the UAVs detection and also guides the generation of the labels to the dataset. Some radar parameters are introduced in Table I. The radar uses a linear frequency modulation (LFM) waveform on a frequency band centered at 35 GHz with a bandwidth of 80 MHz. The pulse repetition frequency (PRF) is set as 32 kHz.

A single received pulse signal is usually not used for detecting the targets because its energy is too low. Hence, the multiple echoes reflected off the targets in a CPI are input into the neural network. During the construction of our dataset, real measure data with three different CPIs (10, 15, and 20 ms) were collected. The pulse-Doppler radar does not receive the echo until the end of signal transmission in each pulse duration, which restricts the minimum radar range. It can be given by the following equation:

$$r_{min} = \frac{cT_e}{2} \quad (1)$$

where T_e is the pulse duration, which is in inverse proportion to the bandwidth of the signal. The pulse-Doppler radar measures the distance of a target by calculating the delay from transmitting the signal to receiving the echo. If the most recent transmitted pulse is before the received echo of the last pulse, it is difficult to make clear which transmitted pulse the received echoes come from. To guarantee that the reflected signal from the target corresponds to the most recent transmitted pulse [36], the maximum unambiguous range is calculated as in the equation below

$$r_{max} = \frac{c}{2f_r}. \quad (2)$$

Considering the maximum unambiguous range and the minimum radar range, the flying distance of the cooperative target (UAV) should be within this range. The maximum phase shift from one pulse to the next that pulse-Doppler radar can measure is π . The phase shift will exceed π if the target moves too far within the time interval between two pulses. That is to say, if the Doppler frequency of the target is not over half PRF $f_d \in (-f_r/2, f_r/2)$, the Doppler frequency shift of the target can be uniquely acquired. Otherwise, it is likely to result in Doppler ambiguity, that is, velocity ambiguity. Here, the Doppler frequency is estimated as $f_d = 2v_r/\lambda$. The maximum unambiguous velocity can be defined as follows:

$$v_{max} = \frac{f_r \lambda}{2}. \quad (3)$$

B. Signal Model

Assume that the complex data matrix after the MF is P_{echo} as follows:

$$P_{echo} = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix} = \begin{bmatrix} e_{11} & e_{12} & \dots & e_{1m} \\ e_{21} & e_{22} & \dots & e_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ e_{n1} & e_{n2} & \dots & e_{nm} \end{bmatrix} \quad (4)$$

where the fast-time dimension is n , and the slow-time dimension is m . Considering multiple pulses in the same fast-time dimension (i.e., range cell), e_n can be written as follows:

$$e_n(m) = I(m) + jQ(m) \quad (5)$$

where j is the imaginary unit. $I(m)$ and $Q(m)$ represent the in-phase component and quadrature-phase component separately, respectively [23]. In practice, the received echo $e_n(m)$ consists of two different sources: the echo $e_{reflect}(m)$ reflected from both the target and the clutter, and the ambient noise $e_{noise}(m)$. Therefore, (5) can be rewritten as follows:

$$e_n(m) = e_{reflect}(m) + e_{noise}(m) \quad (6)$$

where $e_{reflect}(m)$ and $e_{noise}(m)$ vary with the time in the same range cell. The reflected echo $e_{reflect}(m)$ is composed of three components as follows:

$$e_{reflect}(m) = e_{dop}(m) + e_{micro}(m) + e_{clutter}(m) \quad (7)$$

where $e_{dop}(m)$ denotes the received signals caused by the bulk motion of the target, corresponding to the Doppler frequency; $e_{micro}(m)$ is the component caused by the micromotion of

the target, corresponding to the micro-Doppler frequency resulting from the frequency modulation; and $e_{\text{clutter}}(m)$ is the component reflected from the clutter.

For the received signals, $e_{\text{dop}}(m)$ caused by the bulk motion of the target. Suppose that there is a moving target [37] and the instant slant range between the target and the radar within a finite CPI can be approximated by the following equation:

$$R(m) = R_0 + v_0 m + \frac{1}{2} a m^2 \quad (8)$$

where R_0 , v_0 , and a are the initial slant range, radial velocity, and radial acceleration of the target, respectively. Then, the received signal of this target after MF can be expressed as follows:

$$e_{\text{dop}}(n, m) = \alpha \text{sinc} \left[B \left(n - \frac{2R(m)}{c} \right) \right] \times \exp \left(-j4\pi \frac{R(m)}{\lambda} \right) \quad (9)$$

where n is the fast time, and α represents the amplitude of the target echo after ML. As (9) describes, the target envelope shifts with the slow-time m . And the envelope offset is $v_r m + (1/2) a m^2$. The situation that the target envelope offset is kept within one range cell in a finite CPI is only considered, so the RM effect is not discussed in our study. From the phase term in (9), the Doppler frequency of the target can be formulated as follows:

$$f_d(m) = \frac{2}{\lambda} \frac{d(R_0 + v_0 m + \frac{1}{2} a m^2)}{dm} = \frac{2v_0}{\lambda} + \frac{2a}{\lambda} m. \quad (10)$$

Equation (10) is a detailed description of the Doppler frequency caused by the bulk motion of the target within a range cell. Then, the full formulation of the micro-Doppler echo $e_{\text{micro}}(m)$ of the drone is expected to formulate. Suppose that there are U scattering centers (propellers) on the target, and then the overall micro-Doppler frequency corresponding to $e_{\text{micro}}(m)$ can be expressed as follows:

$$e_{\text{micro}}(m) = \sum_{u=1}^U \sigma_u \exp \left(-j2\pi \frac{f_m(m)}{f_s} \right) \quad (11)$$

where f_s is the sampling frequency, and σ_u is the amplitude of the u th scattering center. Then, the micro-Doppler frequency of the target f_m can be formulated as follows:

$$f_m(m) = \frac{2f_c |\psi(m)| \cos(\varphi/2) \cos\beta}{c} \quad (12)$$

where φ is the bistatic angle between the radar and the target, and β is the aspect angle between the bisector direction and $\psi(m)$. $\psi(m) = 2\pi\omega L_h$ is the magnitude of the blade tip velocity of the rotating propeller, ω is the rate of rotation, and L_h is half of the length of the propeller [23]. The received echoes reflected from a UAV are shown in Fig. 1.

As the formulation illustrates, the Doppler frequency and the micro-Doppler frequency of the target are the functions of the variable m . Thus, the UAV detection can be transformed into a binary classification task to judge whether a UAV appears in each range cell or not. We set two categories (UAV present and UAV absent), that is, $\xi = [\xi_0, \xi_1]$. It is well known that regression and classification are the two main tasks of DL-based methods, and the DNN shows higher efficiency

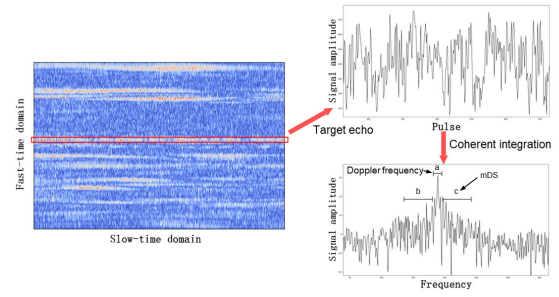


Fig. 1. Echoes reflected from a UAV. The fragment marked with the red rectangle is the received signals reflected from a UAV. After coherent integration, the Doppler spectrum is obtained. Zone “a” is the Doppler frequency, caused by the bulk motion of a UAV; Zone “b” and “c” represent the mDS corresponding to the micromotion of a UAV.

with respect to these tasks. Therefore, the UAV classification problem is defined as follows:

$$\xi = F(P_{\text{echo}}) \quad (13)$$

where $F(\cdot)$ is the multilayer perceptron (MLP) classification function consisting of linear and nonlinear functions. In real, radar works in complex scenes and the pattern of the target is ever-changing. The detection performance of traditional methods relies on the appropriate threshold and accurate estimations of the motion parameters of the target to some extent, whereas DL-based approaches can overcome these problems, since data-driven DL is available for the learning of diverse representations of features.

IV. PROPOSED METHOD

Previous DL-based methods for UAV detection broadly can be divided into two categories: one is to recognize the target's classes and regress position through annotating the bounding box in range-Doppler maps as the ground truth. Although these methods can detect UAVs in a large range, the neural network merely learns the features of the Doppler frequency f_d , but neglects the mDS f_m due to the limited area of the bounding box. Another group is to perform classification for fragment data obtained by a sliding window in the range-Doppler maps or the spectrogram, which considers the mDS f_m of the UAV. However, the precision of localization for the target will affect the result of classification. Different from previous methods, the proposed method transforms UAV detection into a binary classification task to decide whether the target is present in each range cell or not. Fig. 2 shows the framework of the proposed algorithm and the innovations are summarized as follows.

- 1) Instead of the range-Doppler maps, the raw pulse-range data after MF is directly taken as the input. A complex data encoder is designed on the basis of the characteristic of radar complex data. This architecture has a powerful fitting capability, which can substitute for the Doppler processing in the process of traditional methods.
- 2) Considering radar echo is intrinsically temporal data, we present a transformer-based feature extractor with temporal information, which can achieve combination training of pulse-range maps with different CPIs.

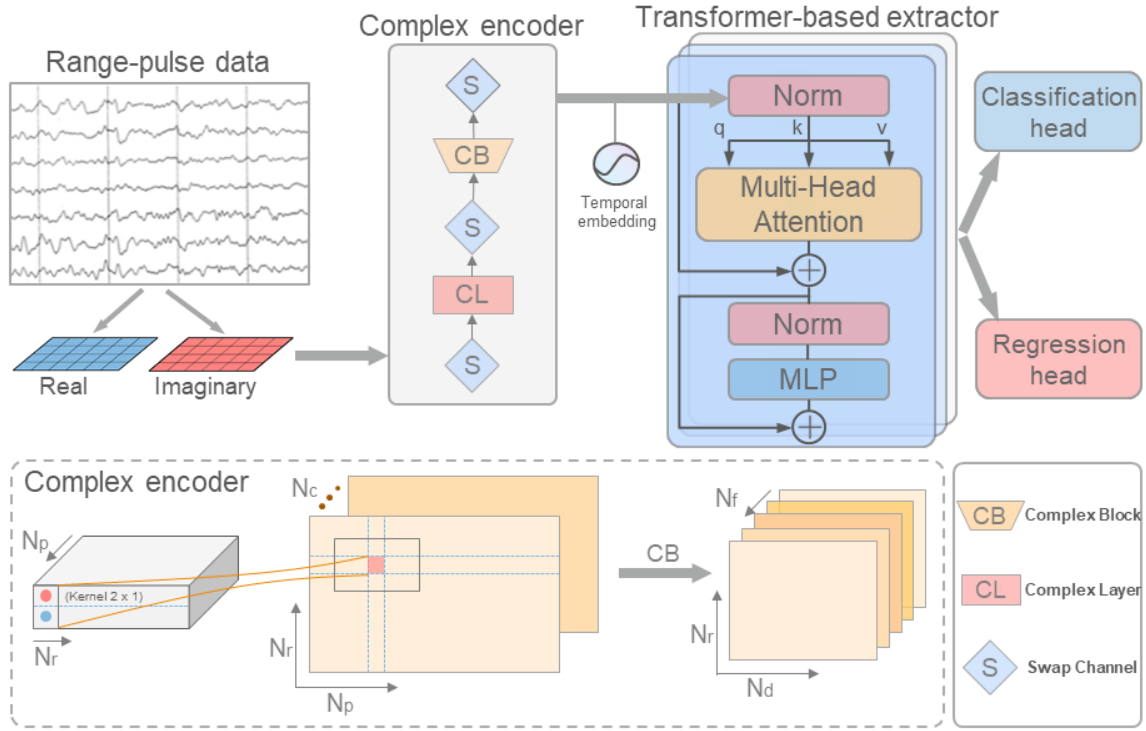


Fig. 2. Framework of the proposed model. The range-pulse data is first sliced into the real and imaginary parts (e.g., the blue patches and the red patches). And then the complex encoder extracts the local feature of radar echoes. After that, the Transformer-based extractor can analyze the global feature of all range cells without the limitation of CPI. This unique design makes us perform combination training with three different CPIs. The number of range cells remains constant during the whole process.

In addition, the attention mechanism is capable of extracting some correlated information between different ranges and Doppler cells. The Transformer-based architecture is more specific to the feature extraction of whole echo data in the same range cell than the local extractor based on CNNs.

- 3) A classification head is arranged to decide whether the target is present in each range cell or not. Meanwhile, a regression head and a loss function for the Doppler frequency shift are adopted to add the Doppler constraints of the entire model, which can further advance the accuracy of classification.

A. Preprocessing of Input Data

Pulse-Doppler radar transmits electromagnetic waves in pulse duration and acquires the echo after the time sampling between two pulses. The received echo is a 1-D discrete sequence in the fast-time domain. Since the energy of a single received pulse signal is quite weak, some researchers usually collect multiple pulses reflected from the targets in a CPI as a 2-D range-Doppler map, which is obtained from noncoherent or coherent integration. We also refer to the range-Doppler map as nonphase values, which can be input into the neural network to extract features without any preprocessing, but the phase of signals is dropped out. Consequently, the nonphase value will damage the temporal-spectral correlation. In contrast, using raw complex numbers (real and imaginary) directly is more suitable for the phase features extraction of the original information contained in the received signals and makes full use of the temporal-spectral correlation as well [38].

The 2-D pulse-range data with three different CPIs is used as input of the network, the dimension of which is $N_r \times N_p$, where N_r denotes the number of range cells within a range gate (fast-time domain) and N_p is the number of pulses in a CPI (slow-time domain). Combination training of pulse-range data with three different CPIs can improve the generalization capability of the model. Here, range-Doppler maps are not taken as the input because the multilayer DNN has a powerful fitting capability enough to replace FFT-common coherent integration methods [39]. Moreover, some scholars think that coherent integration serves as the preprocessing of the training data, which can help to sharpen features. However, using FFT is thought to be equal to a fixed window filter and cannot extract features in a multiscale way. Using multiscale convolution kernels on pulse-range data can extract features, which is equivalent to multiscale cascaded filter groups. Considering the form of pulse-range data is as shown in (5), the complex data is divided into two parts (real and imaginary) and then concatenate them in the third dimension, which will shape a 3-D array as the input of the network. Dimensions of the array are $N_c \times N_r \times N_p$, where N_c denotes the number of complex channels. Fig. 3 shows the operation of the preprocessing.

B. Complex Encoder Structure

In our study, the echo signal after preprocessing is a 3-D array, whose dimensions are $N_c \times N_r \times N_p$, where $N_c = 2$ is initial complex channels. Although the complex data is divided into two channels (real channel and imaginary channel), it represents one value in the same N_c . Therefore, when performing the operations of convolution, we should

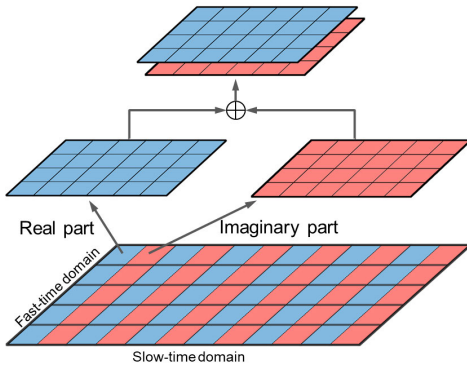


Fig. 3. Operation of the preprocessing. The range-pulse data is first sliced into real and imaginary parts. $\oplus$ represents the concatenation operation.

avoid mixing the real part and imaginary part of the different cells, lest the convolution wreck the intrinsic representation of the complex number. To this end, a special structure encoder is designed to exploit time–frequency information about radar echo signals. Specifications of the encoder are summarized in Table III. First, a new dimension is expanded into the pulse-range array. Dimensions of it are $N_p \times 1 \times N_c \times N_r$. The representation of high-dimensional features ($N_p \times N_c \times 1 \times N_r$) to the complex data is gained in each complex channel dimension. In the above process, the size of the convolution kernel corresponding to the range cell is 1, which ensures that the computation of complex data is not confused. Second, the features of Doppler frequency are filtered and extracted in multiple Doppler cells. The feature array is permuted ($1 \times N_c \times N_p \times N_r$) to be input to convolution layers and further be extracted the features of Doppler frequency in detail. Finally, a high-dimensional feature cube is calculated, where the number of range cells is unchanged. Dimension operations like expanding and permuting are to be convenient to calculate the data feature by using convolution operations.

Notice that the size of the range cell remains unchanged in feature extraction of the Doppler frequency, for the purpose of classification for each range cell. The size of the convolution kernel corresponding to the range cell is set as 3, which can extract the features of adjacent range cells. This convolution kernel can accumulate the Doppler frequency features of adjacent range cells to overcome measurement errors and sampling errors in a radar system. In addition, since the convolution operation is similar to the filtering, using multiscale convolution kernels in the Doppler cells has the same effect as the windowed filtering groups, which can dig into the time–frequency feature of the received signals. The high-dimension features are more beneficial to the representation of the mDS than the low-dimensional features. It has better linear separability for downstream tasks, such as regression and classification [40].

C. Design of the Transformer-Based Extractor

A complex encoder structure maps the complex data into high-dimensional space. Although this structure extracts time–frequency features in multiple scales, it merely obtains

local features. The energy of signals can be increased by collecting multiple pulse signals, which is also available for DL-based methods. Extracting the features of whole Doppler cells in a CPI is good for the analysis of the Doppler features, especially for mDS with weak energy. The correlation and difference between range cells can be calculated so that the range cells perform more accurate classifications. In addition, the global features always reflect ambient noise levels, ground clutter, and other information, so it is necessary to extract and analyze global time–frequency features [38].

In previous CNN-based methods, the global features are extracted by increasing the depth of the network and enlarging the receptive field. The deep features are full of semantic information, but it is detrimental to the small target detection in that the small target usually only occupies a few cells in the range-Doppler maps [5]. Moreover, most of the well-applied pipelines for object detection demand a fixed size of the input. To realize UAV detection for radar echo with different CPIs, a Transformer-based architecture is designed specially, which already has a popular backbone to extract the features in the field of CV and NLP. It has been verified in most tasks that the transformer feature extraction capability is superior to that of the CNN-based networks.

Patch embedding: A 3-D feature cube is obtained after the complex encoder structure. The MLP is used to get a range-based feature patch ($N_r \times N_d \times N_f$), where N_d is the number of the Doppler domain, and N_f denotes the length of the feature vector. Some scholars think that the attention mechanism cannot capture the time-series information [41], which the radar echoes include. To compensate, the temporal embedding (TE) with learnable parameters is introduced into the algorithm and then patch embedding can be expressed as follows:

$$x = \text{MLP}([f_1, f_2, \dots, f_{N_r}]) + E_{te} \quad (14)$$

where f_i is the feature vector of the i th range cell, MLP denotes the MLP, and E_{te} is TE that can express the received signal sequence with prior information in the fast-time domain.

Multihead attention: The received radar signals have the time–frequency coherent relation in both fast-time and slow-time domains, so single pulses hardly reflect the target features. The self-attention mechanism is available for calculating the time–frequency coherent relation of echoes, which can have a sufficient and complete representation of the Doppler features and mDS of the target. Self-attention will map the patch embedding into three new vectors with the same dimensions: Query ($q \in R^{(N_r \times N_d \times N_f)}$), Key ($k \in R^{(N_r \times N_d \times N_f)}$), and Value ($v \in R^{(N_r \times N_d \times N_f)}$). The key, query, and value are computed as linear projections of deep features, which are formulated as follows:

$$\begin{bmatrix} q \\ k \\ v \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} \cdot x + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad (15)$$

where w and b denote learnable weights. To enhance the representation capability of the network, our architecture uses multihead attention where the original feature vectors are duplicated H times and then perform the attention function in

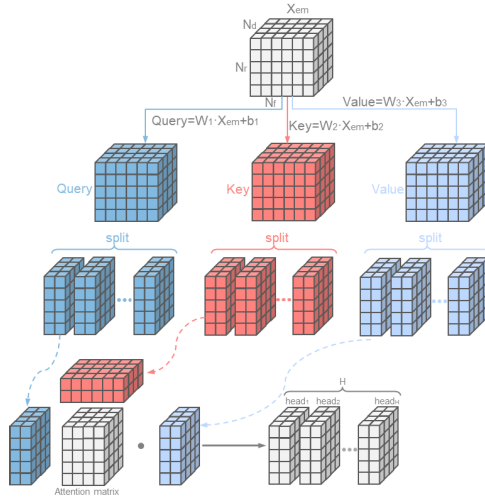


Fig. 4. Process of multihead attention. The query, key, and value are computed as linear projections of the range-based feature patch ($N_r \times N_d \times N_f$) and then are split into H parts. $\cdot$ is a symbol of matrix multiplication.

parallel. The dimensions of q , k , and v is adjusted to reduce the computing cost: $N_f = N_h/H$, that is, q , k , and v are both split into H parts. Overall multihead attention is defined as follows:

$$x' = \parallel_{h=1}^H \text{softmax} \left(\frac{q^{(h)} \cdot k^{(h)\top}}{\sqrt{N_h}} \right) \cdot v^{(h)} \quad (16)$$

where $\parallel$ denotes the concatenation operation, and $\text{softmax}(\cdot)$ is the softmax function. The whole process of multihead attention is presented in Fig. 4. The Transformer-based architecture is composed of alternating layers of multiheaded self-attention and MLP blocks. Layernorm (LN) is applied before every block, and residual connections after every block [42]. It can help release gradient vanishing and explosion during model training. Multiple Transformer-based architectures in series can sufficiently extract the global time–frequency features of the received signals to further increase the difference between range cells.

D. Loss Function

The number of range cells remains constant (N_r) in the architecture ahead. The UAV detection is taken as the classification task in each range cell where the position of the target is obtained directly. In the process of recording radar echoes, not only can we record the distance of a UAV, but also obtain the radial flying speed, which is key ground truth for our training. According to the Doppler effect, the radial speed of the target is a remarkable signature, corresponding to the Doppler frequency. Therefore, a tight coupling way is used during the classification, where the Doppler position of the target in Doppler cells is regressed simultaneously. The classification and regression heads are connected to transformer blocks. Thus, the dimensions of the output are $N_r \times 3$. Notice that the two-class values with target and no-target outputs and a Doppler regression value are considered in this task. The cross-entropy loss is our classification loss function, which is

given by the following equation:

$$L_{\text{cls}} = \frac{1}{N_r} \sum_i^{N_r} -[y_i^r \cdot \log(p_i^r) + (1 - y_i^r) \cdot \log(1 - p_i^r)] \quad (17)$$

where y_i^r denotes the label of the i th range cell, and p_i^r is the probability of the target occurring in the i th range cell. The prediction of the Doppler frequency position of the target can be taken as a regression problem. If the target is present in a range cell, the Doppler frequency loss will be calculated, and the modified L1-loss is defined as follows:

$$L_{\text{dop}} = \frac{1}{n} \sum_i^{N_r} y_i^r \cdot |y_i^d - p_i^d| \quad (18)$$

where y_i^d is a symbol for the Doppler frequency position of the i th range cell, p_i^d denotes the predicted value of the Doppler frequency position, and n is the number of range cells with the target. The overall loss function can be written as follows:

$$L_{\text{total}} = L_{\text{cls}} + \lambda L_{\text{dop}} \quad (19)$$

where λ is the weight factor that balances the contribution of the two loss terms. Coupled heads are adopted at the end of the network, predicting the category and the Doppler frequency value instead of decoupled heads. One reason is to reduce model parameters; another is that the Doppler frequency is one of the obvious signatures to judge whether the target is present or not. Accordingly, coupled heads can better constrain the classification of range cells at the same time.

V. DATASET CONSTRUCTION

In related works, most methods were evaluated on simulated datasets. Although they can add noise to source data and simulate some UAVs with diverse signal-to-noise ratios (SNR), targets are generated by the theoretical model of a flying target, which is not close to the real situation. However, some works released their real measured datasets. Roldan et al. [4] issued a novel trainable dataset with more than 17 000 samples of drones, cars, and people, acquired in real outdoor scenarios, but the collected data have been trimmed around the detected object, that is, the position of the target has been known in advance. A micro-Doppler recognition dataset was issued in [43], which is available for the classification of four types of objects (vehicle, walking person, bicycle, UAV). To evaluate the UAV detection performance of the proposed method, a pulse-Doppler radar is used to build a UAV detection dataset available for end-to-end training. The performance parameters of the radar are introduced in Table I. Our cooperative target is a fixed-wing drone, which is 2–4 km away from the radar, and the radial velocity of which is 30–60 m/s. The drone is equipped with a global positioning system (GPS) module, and the state information (distance and velocity) is received at each interval of 10 ms.

Unlike image-related tasks, DL-based methods have yet to be widespread in tasks about radar raw data. There are three main reasons: The form of radar raw signals is too abstract to intuitively understand. Data augmentation and other training skills about radar raw signals are extremely lacking.

The annotation of radar data demands professional knowledge in the field of radar signal processing. Therefore, our work offers a means of converting UAV state data (distance and velocity) into trainable labels available for DL-based methods and an approach that can generate target signals, which are both beneficial to the improvement of generalization.

A. Target Annotation

Since UAV state data (distance and velocity) of the cooperative target is not available for the training of DL-based algorithms, it is necessary to convert the distance and radial velocity into trainable labels. According to the loss function, the ground truth of the category and the Doppler frequency shift position of all range cells is needed to be labeled. Assume that the received distance information between radar and UAV is d_u m, the range cells that have the target can be expressed as follows:

$$R_{\text{pos}} = \text{round}\left(\frac{d_u - r_{\text{min}}}{\Delta r}\right) \quad (20)$$

where $\text{round}(\cdot)$ is the nearest rounding function. The range cells that have the target are set as 1, others are set as 0. The Doppler frequency caused by radial speed is vital to ground truth for UAV detection, and this constraint can improve the classification accuracy of range cells. Likewise, suppose that the radial flying velocity is v_u m/s and then the position of Doppler frequency is given by the following equation:

$$D_{\text{pos}} = \begin{cases} \text{round}\left(\left(1 - \left|\frac{v_u}{v_{\text{max}}}\right|\right) \cdot \frac{T_{\text{cpi}}}{T_r}\right) & \text{if } v_u < 0 \\ \text{round}\left(\frac{v_u}{v_{\text{max}}} \cdot \frac{T_{\text{cpi}}}{T_r}\right) & \text{if } v_u \geq 0 \end{cases} \quad (21)$$

where T_{cpi} is CPI. Real measure data with three different CPIs (10, 15, and 20 ms) were recorded during the construction of our dataset.

B. Sample Frequency Shift Augmentation

The UAV only occupies a few cells in the range-Doppler domain, and the received signals reflected from drones are weak sometimes. The above factors have an adverse influence on UAV detection. It is necessary for small object detection tasks to add learnable samples, which has been validated in [44]. In addition, there is only one cooperative target (UAV) to build the dataset, and the lack of sample diversity in the echoes is an intractable problem as well. To overcome these problems, we present the SFSA to generate target signals, which can improve the performance and generalization of the network.

In the range-pulse data, the sample value of the received signals reflected from the target is defined as $x_a(n)$, where a is the a th range cell, $n = 0, 1, \dots, N-1$, and N is the number of pulses in a CPI. First, we perform the coherent integration by discrete Fourier transform (DFT), and the resulting coherent integration signal is

$$X_a(k) = \sum_{n=0}^{N-1} x_a(n) e^{-j \frac{2\pi}{N} kn}. \quad (22)$$

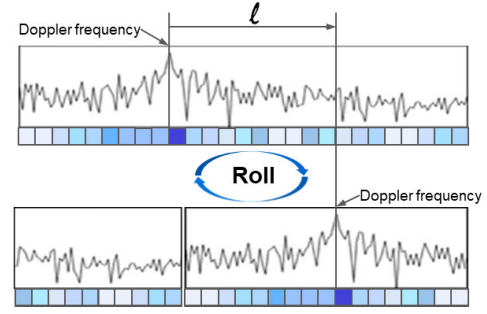


Fig. 5. Illustration of SFSA. The dark blue grid represents the position of the Doppler frequency in the Doppler domain. The received echoes rolled l Doppler cells.

Inspired by time shift augmentation in audio signal processing [45], then assume that the Doppler frequency of the target shifts l unit, the rolling shift in the Doppler domain can be expressed as follows:

$$\begin{aligned} X_a(k+l) &= \sum_{n=0}^{N-1} x_a(n) e^{-j \frac{2\pi}{N} (k+l)n} \\ &= \sum_{n=0}^{N-1} x_a(n) e^{-j \frac{2\pi}{N} kn} \cdot e^{-j \frac{2\pi}{N} ln}. \end{aligned} \quad (23)$$

Suppose that there is no target in the b th range cell, and the received signals is defined as $x_b(n)$, and then add to (23), that is,

$$\begin{aligned} Y_b(k) &= \sum_{n=0}^{N-1} x_b(n) e^{-j \frac{2\pi}{N} kn} \\ &+ \alpha \sum_{n=0}^{N-1} x_a(n) e^{-j \frac{2\pi}{N} kn} \cdot e^{-j \frac{2\pi}{N} ln} \end{aligned} \quad (24)$$

where α denotes the signal amplitude. Finally, the augmentation signal in the b th range cell after the inverse DFT (IDFT) is

$$\text{IDFT}[Y_b(k)] = y_b(n) = x_b(n) + x_a(n) e^{-j \frac{2\pi}{N} ln} \quad (25)$$

where $y_b(n)$ is the target signal in the time domain through sample frequency shift augmentation, that is, generating a target in the b th range cell. Fig. 5 illustrates the operation of SFSA concretely. For multitarget detection, the number of targets is chosen randomly between 3 and 7, which can improve the performance and generalization of the network.

VI. EXPERIMENTAL EVALUATION

A. Experimental Setup

The proposed algorithm is an end-to-end trainable pipeline. The specifications of the complex encoder structure in our model have been listed in Table II. To prevent overfitting in the MLP blocks, the dropout rate is set as 0.1. Six layers of the Transformer block with eight heads are used in the algorithm. To balance the contribution of the two loss terms, the weight factor is set as 1.2. The adaptive moment estimation (Adam) optimization is applied to minimize the loss function. We adopt it with an initial learning rate of 0.0001, betas 0.9, 0.999. The learning rate will be decreased at a factor of 0.5 every 16 epochs. PyTorch is used for model implementation and our model is trained on an NVIDIA RTX 3090 GPU for 145 epochs.

To make a fair and reasonable comparison, some metrics are computed to evaluate the performance of the considered

TABLE II
SPECIFICATIONS OF COMPLEX ENCODER STRUCTURE

Layer Name	Convolution Kernels	Stride	Padding
Complex layer	$2 \times 1@8$	(1,1)	(0,0)
Conv1	$32 \times 3@32$	(3,1)	(0,1)
Conv2	$16 \times 3@64$	(3,1)	(0,1)
Conv3	$8 \times 3@128$	(3,1)	(0,1)

methods. For radar UAV detection, the target detection is taken as a two-class classification problem for each range cell in our work. We denote the number of range cells occupied by UAV samples being correctly classified and falsely classified as n_{tp} (true positive) and n_{fn} (false negative), and the number of range cells occupied by non-UAV samples being correctly classified and falsely classified as n_{tn} (true negative) and n_{fp} (false positive), separately. The probability of detection (P_d) measures the ratio of correctly predicted target cells over all target cells. P_d can be expressed as follows:

$$P_d = \frac{n_{tp}}{n_{tp} + n_{fn}}. \quad (26)$$

The probability of a false alarm (P_f) measures the ratio of falsely predicted target cells over all range cells. P_f is defined as follows:

$$P_f = \frac{n_{fp}}{n_{tn} + n_{fp}}. \quad (27)$$

F1 score is defined as the harmonic mean of precision and recall and can balance both of them. The F1 score can be written as follows:

$$F1 = \frac{2n_{tp}}{2n_{tp} + n_{fp} + n_{fn}}. \quad (28)$$

In the proposed loss functions, the Doppler frequency shift loss is designed particularly. To evaluate the predicted error of the Doppler frequency ϵ , a metric [average Doppler frequency error (ADFE)] is set in this article, which is defined as follows:

$$E_\xi = \frac{n_{|\epsilon| < \xi \cdot M}}{n_{all}} \quad (29)$$

where ξ denotes the Doppler frequency error as a percentage of whole Doppler cells, and M is the number of whole Doppler cells. Our dataset includes pulse-range data with three different CPIs, so the number of M is different. $n_{|\epsilon| < \xi \cdot M}$ is the number of range cells whose predicted Doppler error is less than $\xi \cdot M$, and n_{all} denotes the number of range cells with the target. Three proportions ($E_{1\%}$, $E_{2\%}$, $E_{5\%}$) are set to evaluate the performance of our method.

B. Analysis of Dataset

The effectiveness of DL models highly depends on the generalization of models and the availability of high-quantity data [38]. To improve the robustness of our algorithm, diverse situations, including different positions, SNRs, and flying speeds of targets, have been considered in our labeled radar dataset. In total, the real measured dataset has 3135 data frames, including 2508 examples for training and 627 examples for testing. Although the input of our model is pulse-range data, UAVs are submerged in the background clutter,

which is not convenient for visual analysis. Therefore, the range-Doppler maps after coherent integration are used to analyze the dataset.

In real applications, the range-Doppler maps with different CPIs are adopted for flying target detection to ensure that the moving distance of the targets within a CPI cannot cross one range cell, so we built the dataset with three different CPIs to study the effect of CPIs on UAV detection. The range-Doppler with three different CPIs (10, 15, and 20 ms) are shown in Fig. 6. It can be seen that the longer CPI is, the more obvious the energy of the target is. There are four targets in the shown range-Doppler spectrum. The SNRs of target 1 and target 3 are low, which is not clear in Fig. 6(c). The SNR of target 2 is higher than target 3, so the Doppler modulation is more clear. Our dataset has plenty of UAVs with SNRs from -10 to 20 dB, which keeps the diversity of samples to some extent. The detail of energy accumulation is presented in Fig. 7. The more strong energy accumulation close to the zero frequency, that is, ground clutter, is shown in Fig. 7(a). The energy accumulation of the position of targets with different SNRs is shown in Fig. 7(b) and (c), as we described in Section II-B, the echoes reflected from the target can be divided into two parts: e_{dop} corresponding to the Doppler frequency; e_{micro} corresponding to the micro-Doppler frequency resulting from the Doppler frequency modulation. In Fig. 7(d), the clutter occurs nearby the zero frequency where the Doppler frequency modulation does not exist in spite of clear energy accumulation near the area of the zero frequency. This also proves that the core idea of the proposed method is reasonable—the Doppler frequency and mDS need to be considered simultaneously. In addition, to test the performance of our method for detecting UAVs with hovering or small radial velocity, the targets close to zero frequency (<8 m/s), called the zero-frequency target, are generated as well. In Fig. 8, the energy of the target is almost submerged in the ground clutter.

C. Comparison to Previous Approaches

In this section, five different UAV detection methods are used for comparison to shed light on the advanced performance of our algorithm. A CFAR is indispensable in traditional radar target detection methods, but DL-based methods, including our algorithm, do not have the detection window and the relationship between the false alarm and the detection threshold. Therefore, the same metric setting was adopted as in [5], [30], and [32], that is, testing on the whole test set under different SNRs. CA-CFAR [18] and OS-CFAR [20], whose size of a sliding window is 15 and guard cells are 3, are conventional methods which are widespread in radar target detection. Another is a DL-based method proposed by [30], which handles the detection problem as a classification task between target and clutter. The fourth approach is based on object detection in images [5], whose core idea is inspired by Faster RCNN. As described in Section II, using micro-Doppler-based methods for UAV detection is another popular solution, so [34] is enumerated for comparison. All the above methods are reproduced by us and then evaluated on our dataset. Table III shows the detection results of the six methods on our dataset. The average Doppler frequency shift

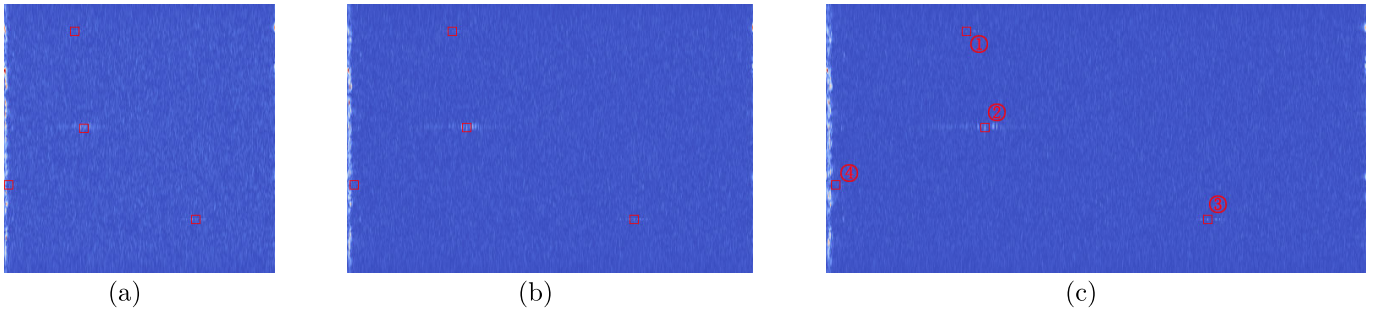


Fig. 6. Range-Doppler maps with different CPIs. There are four targets marked with red bounding rectangles in the range-Doppler map in total. Target 2 is more energy accumulation than others, and the longer CPI is, the more energy the target is: (a) CPI=10 ms; (b) CPI=15 ms; and (c) CPI=20 ms.

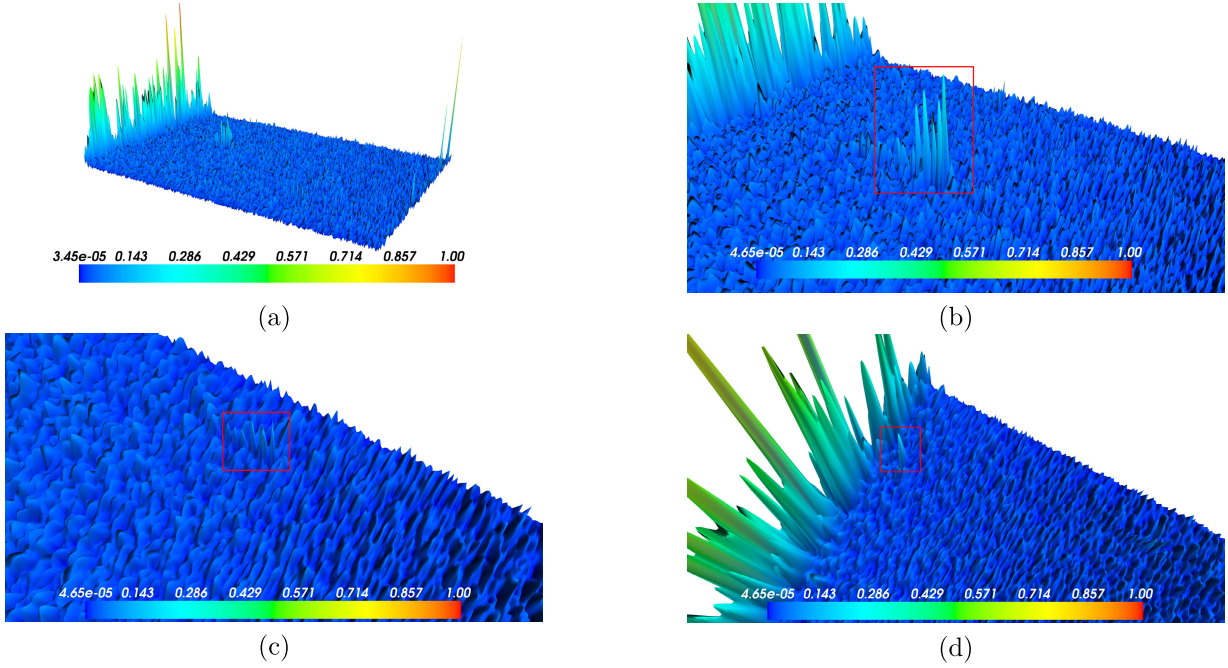


Fig. 7. Range-Doppler map after coherent integration. Our dataset has plenty of UAVs with SNRs from -10 to 20 dB. The SNR of target 2 is higher than target 1. In addition, it is difficult to distinguish UAVs with low SNRs and clutter just according to the energy accumulation. (a) Range-Doppler map visualization. (b) Target 2 with higher SNR. (c) Target 1 with lower SNR. (d) Ground clutter.

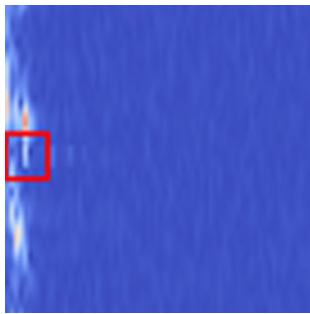


Fig. 8. Demo of the zero-frequency target. The Doppler frequency shift of the UAV is close to zero. There are lots of ground clutter in the zero frequency area, and the echoes are almost submerged in the ground clutter.

error is not suited for evaluating the results in five other methods, so our experiments merely use three kinds of metrics related to classification to compare the performance. The P_d , P_f , and $F1$ scores of the proposed method are 0.972, 0.0034,

and 0.976, respectively, which are all superior to others. Wang et al. [30] have a better performance than [5], but the disadvantages of methods based on the sliding window are obvious. The position of targets cannot be located accurately, and the error of target location will pose an undesirable impact on results. The proportion of the Doppler frequency error within 1% is 0.554, and the Doppler frequency error within 5% accounts for 93.8%. The proposed method achieves promising results for the prediction of the Doppler frequency shift. The other three methods do not take ADFE as the metric to evaluate their algorithm.

D. Performance Evaluation on Noise Robustness

To evaluate that our method has ideal robustness to drones with diverse SNR levels, UAV examples with different SNRs are generated by setting different the energy of target signals. Fig. 7(b) and (c) shows the power of targets with different SNRs. The results of the proposed method under different SNRs are shown in Table IV. The SNR varies from

TABLE III
COMPARISON OF SIX METHODS ON OUR DATASET

Methods	P_d	P_f	F1 score	$E_1\%$	$E_2\%$	$E_5\%$
CA-CFAR [18]	0.856	0.0252	0.712	-	-	-
OS-CFAR [20]	0.864	0.0276	0.731	-	-	-
Wang and Tang et al. [30]	0.892	0.0149	0.854	-	-	-
Wang and Tian et al. [5]	0.886	0.0167	0.821	-	-	-
Le and Doan et al. [34]	0.895	0.0115	0.844	-	-	-
Proposed method	0.972	0.0034	0.976	0.554	0.774	0.938

TABLE IV
COMPARISON OF THE PROPOSED METHOD UNDER DIFFERENT SNRS

SNR(dB)	P_d	P_f	F1 score
-10 – -5	0.9	0.163	0.858
-5 – 0	0.973	0.0274	0.926
0 – 5	0.981	0.0143	0.934
5 – 10	0.998	0.0101	0.991
10 – 15	0.997	0.0071	0.993
15 – 20	0.999	0.0001	0.998

TABLE V
EVALUATION FOR THE ZERO-FREQUENCY TARGET

Target type	P_d	P_f	F1 score
zero-frequency UAVs	0.997	0.0479	0.968
UAVs(≥ 8 m/s)	0.975	0.0029	0.98

–10 to 20 dB, which can help us validate the antinoise capacity of our method. P_d and F1 score are close to 1, and P_f is 0, when the SNR is greater than 15 dB, so the proposed approach achieves a pretty good performance. When the SNR declines, P_f gradually increases, which indicates that the energy of targets is similar to the noise, and some noise is falsely detected as the targets. The F1 score also decreases at the same time after false detection and missed detection has arisen. Therefore, the proposed algorithm has strong robustness for targets with different SNR levels. Although different SNRs can directly make a difference to the representation of the Doppler frequency shift, to some extent, the architecture of our model will compensate for this shortage due to extracting the mDS in all Doppler cells.

E. Performance Evaluation for Zero-Frequency Target

The Doppler frequency of hovering or tangential flying UAVs is close to zero frequency, however, the zero-frequency area is often mixed with plenty of ground clutter and various noise, which has an intractable influence on detecting detect drones with low flying speed from the background. To illustrate the performance of our method on zero-frequency targets, Table V presents the detection results of these targets. The UAV with a speed of less than 8 m/s is defined as the zero-frequency target. P_f has a rise, and a slight decline occurs in the F1 score compared to Table III. This illustrates that our method achieves good detection outcomes for UAVs with low speeds.

F. Performance Evaluation by Varying CPI

In practice, the CPI of radar echoes is usually adjusted to realize more ideal detection results, so it is necessary to dig

into the influence of the varying CPI on DL-based methods. For the above-mentioned reason, the present method built real measure data with three different CPI (10, 15, and 20 ms) and skillfully designed the transformer-based model to achieve combination training of pulse-range maps with different CPI. Table VI shows many groups of the outcomes of our test. Compared to the training on the dataset with a single CPI, the results of the training on the dataset with CPIs of 10 and 15 ms are more ideal, and even the model training on the dataset with three different CPIs has the most excellent performance during our tests. This sheds light on an essential conclusion: training on the dataset containing different CPIs has a positive effect on the detection performance because varying the CPI of radar data like a type of augmentation skill will improve the generalization of the model.

The model training and testing on the 10-ms CPI gets the poorest performance, which is affected by the noise and clutter. Too short CPI will cause poor accumulation of the target echoes to detect weak targets accurately. In contrast, the longer the CPI of the test set is, the better outcomes our method gets. When we trained our model on the dataset with three CPIs and tested on the CPI of 15 ms, P_d is at the peak (0.978), the minimum P_f is 0.0022, F1 score is 0.983, and ADFE is also ideal. As seen in Table VI, our model that is performed combination training of the dataset with three different CPI has better generalization.

G. Performance Evaluation on the Sample Frequency Shift Augmentation

To increase the diversity of the sample, we propose an SFSA means to generate target signals, which is a kind of training skill for radar echoes. The proposed models are trained on the two datasets with SFSA and without SFSA, respectively. The evaluation results testing our dataset are shown in Table VII. The values of P_d do not have an obvious change, but there is a tremendous decline for P_f and the F1 score increases from 0.505 to 0.976, when using SFSA on our dataset. In addition, the prediction of the Doppler frequency has less error compared to the dataset without SFSA. Hence, the SFSA acts on improving the model performance and generalization.

H. Ablation Experiments

To evaluate the effectiveness of our design decisions, ablation studies can help us better understand how the components of our model affect the overall performance of the proposed algorithm. Table VIII enumerates some outcomes of ablation experiments by changing different components of our model.

TABLE VI
COMPARISON OF THE DATASET FOR VARYING CPI

Training set			Test set			P_d	P_f	F1 score	$E_{1\%}$	$E_{2\%}$	$E_{5\%}$
10ms	15ms	20ms	10ms	15ms	20ms						
✓			✓			0.784	0.0345	0.748	0.172	0.276	0.417
✓			✓	✓	✓	0.885	0.0277	0.763	0.122	0.231	0.419
	✓			✓		0.942	0.0102	0.924	0.236	0.449	0.705
	✓		✓	✓	✓	0.929	0.0104	0.924	0.229	0.464	0.721
		✓			✓	0.939	0.0096	0.928	0.264	0.445	0.656
		✓	✓	✓	✓	0.808	0.0229	0.855	0.246	0.422	0.636
✓	✓		✓	✓		0.946	0.0077	0.946	0.399	0.536	0.745
✓	✓		✓	✓	✓	0.952	0.0067	0.952	0.398	0.555	0.764
✓	✓	✓	✓			0.961	0.0051	0.965	0.519	0.718	0.912
✓	✓	✓		✓		0.977	0.0027	0.981	0.569	0.799	0.955
✓	✓	✓			✓	0.978	0.0022	0.983	0.577	0.814	0.951
✓	✓	✓	✓	✓	✓	0.972	0.0034	0.976	0.554	0.774	0.938

TABLE VII
EVALUATION FOR SFSA

Dataset	P_d	P_f	F1 score	$E_{1\%}$	$E_{2\%}$	$E_{5\%}$
w/o SFSA	0.971	0.0471	0.505	0.2303	0.339	0.379
SFSA	0.972	0.0034	0.976	0.5536	0.7743	0.9382

TABLE VIII
ABLATION STUDIES OF THE PROPOSED METHOD

Model	P_d	P_f	F1 score	$E_{1\%}$	$E_{2\%}$	$E_{5\%}$
w/o multi-head	0.948	0.0058	0.961	0.436	0.5907	0.8204
A layer of transformer block	0.962	0.0048	0.966	0.3266	0.5212	0.8234
w/o TE	0.952	0.0057	0.943	0.6026	0.751	0.9127
w/o the Doppler frequency loss	0.956	0.0056	0.954	0.0045	0.011	0.0311
CNN-based extractor	0.934	0.0096	0.922	0.496	0.6489	0.8349
Proposed model	0.972	0.0034	0.976	0.5536	0.7743	0.9382

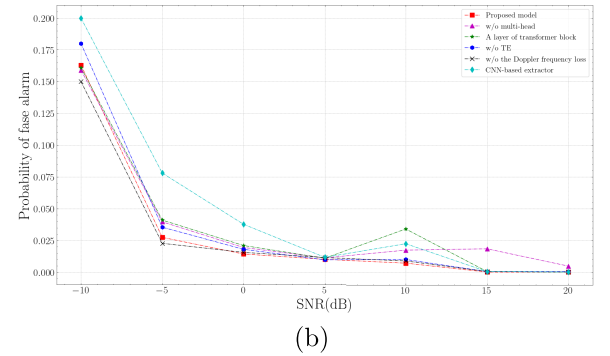
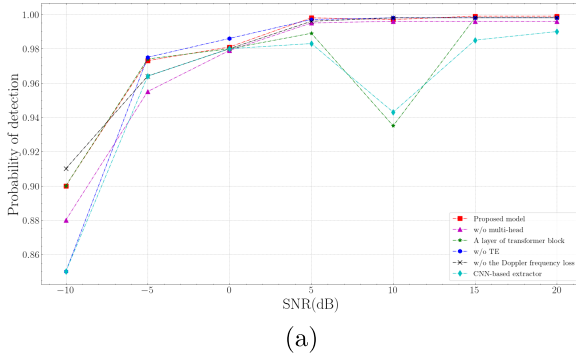


Fig. 9. Detection performance of the ablation studies under different SNRs. (a) P_d of the ablation studies. (b) P_f of the ablation studies.

Moreover, the results under different SNRs are presented in Fig. 9. The design of the multihead attention module has slight gains in respect of the probability of detection and false alarm, but there is a remarkable increase for the ADFE. Our model with a layer of transformer block is in a similar situation. The introduction of the TE makes the performance of our model acquire improvement to some extent. Adding the Doppler frequency loss lets our model have the capability of predicting the Doppler frequency shift, that is, there is a dramatic increase in the ADFE. Besides that, our design makes the detection results more stable.

VII. CONCLUSION

In this work, considering radar echo characteristics, a novel Transformer-based architecture available for the end-to-end training is proposed, which can extract the Doppler frequency shift feature and the mDS of a UAV simultaneously. As Doppler processing (coherent integration) can be seen as convolution operations, the raw pulse-range data instead of range-Doppler maps after coherent integration is directly taken as the input. The complex encoder and the Transformer-based extractor are designed to achieve combination training of pulse-range maps with different CPIs. To this end, detailed

experiments shed light on the influence of different CPIs on DL-based methods. In addition, this article provides an approach to annotate radar data available for DL-based algorithms and presents the SFSA to perform data augmentation. Experiments demonstrate that the proposed method achieved a more robust performance than previous algorithms. Finally, a real measured dataset collected with a pulse-Doppler radar is built to facilitate the development of DL-based methods for UAV detection.

In the future, we will focus on realizing multiclass target detection using raw radar signals, such as birds, quadcopters, and so on. Building multiclass datasets could boost the development of DL-based methods for radar target detection.

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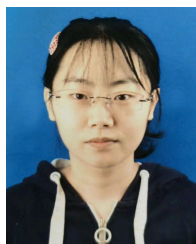
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